

The Two Halves of One Product: The Jordan/Malcev Split of the Octonions and the Measurement Channel

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Abstract. This series has developed two strands from the octonions: a Jordan strand — $J_3(\mathbb{O})$, its Peirce structure, frame relativity, the JBW context — and a Malcev strand — algebraic R-flux, the contraction tower, the squeeze-thermal state, the type III₁ verdict. This short note proves the two strands are the symmetric and antisymmetric halves of *one* product. On imaginary octonions the split is exact and extreme: the symmetrized (Jordan) half of any product is *purely scalar*, equal to $-\langle a, b \rangle e_0$, while the antisymmetrized (Malcev) half is *purely imaginary* — verified to machine precision. The full symmetrized algebra $\mathbb{O}^+$ satisfies the Jordan identity, proved here via Artin's theorem rather than cited. The scalar emission of the Jordan half is identified as the *measurement channel* — the algebraic locus where relations between directions register as real numbers — and three formalization rules are fixed to keep its treatment consistent with the series' type III₁ architecture: evaluation, never projection; pairing as the template of states, with the geometry-only state (the trace) forbidden; and outcome-production explicitly left unformalized. An operator-algebraic mirror is noted: Alfsen-Shultz reconstruct the associative product as Jordan product plus dynamical correspondence, $ab = a \circ b - i(a \star b)$ — the same split, one stratum up.

1. One product, two strands

For pure imaginary octonions a, b , the product is $ab = -\langle a, b \rangle e_0 + a \times b$.

Symmetrizing and antisymmetrizing:

Proposition 1 (verified). On $\text{Im}(\mathbb{O})$: the Jordan half $\frac{1}{2}(ab + ba) = -\langle a, b \rangle e_0$ is *purely scalar*, and the Malcev half $\frac{1}{2}(ab - ba) = a \times b$ is *purely imaginary*. Verified across 300 random pairs: imaginary residue of the symmetric half 0 exactly; deviation of its scalar from $-\langle a, b \rangle$ at 1.8×10^{-15} ; real residue of the antisymmetric half 0 exactly. In particular the self-relation is the scale: $a \circ a = -|a|^2 e_0$ (verified, 3.6×10^{-15}).

The antisymmetric half is the non-Lie Malcev algebra proven unique in the Hurwitz chain — the origin of the R-flux strand, the contraction tower, and ultimately the squeeze-thermal state. The symmetric half is where the Jordan strand begins: measurement values, since Jordan-von Neumann-Wigner (1934), live in Jordan algebras. The series' two strands were never two subjects; they are the two halves of one multiplication.

2. The symmetrized octonions form a genuine Jordan algebra

Proposition 2. $\mathbb{O}^+ = (\mathbb{O}, \circ)$ with $a \circ b = \frac{1}{2}(ab + ba)$ satisfies the Jordan identity $(x \circ y) \circ (x \circ x) = x \circ (y \circ (x \circ x))$.

Proof. The identity involves only the two generators x, y . By Artin's theorem, the

subalgebra of $\mathbb{O}$ generated by any two elements is associative; on an associative algebra the symmetrized product is special Jordan, so the identity holds there — hence in $\mathbb{O}^+$. ■
 Verified independently: maximal residual 5.7×10^{-14} across 200 random full-octonion pairs.

Two structural remarks. First, the series' non-closure leitmotif recurs here in miniature: $\text{Im}(\mathbb{O})$ is *not closed* under $\circ$ — its products land in $\mathbb{R}e_0$, outside the subspace — and this emission is the channel of interest, not a defect. Second, a precision that matters for scope: restricted to imaginary elements the Jordan half is degenerate as an algebra (every product a scalar); the *rich* Jordan structure of this series lives in $\mathbb{O}^+$ and above all in $J_3(\mathbb{O})$. What the imaginary restriction isolates is not an interesting algebra but an interesting *channel*.

3. Two scalar emissions: the metric and $\hbar$

Relations between imaginary directions produce real numbers through exactly two channels, one per half. The symmetric channel emits the *metric* data: overlaps $-\langle a, b \rangle$ and scales $-|a|^2$, landing on the real axis. Note that on a rigid orthogonal basis this channel is silent (orthogonal units emit zero); it fills precisely when directions vary continuously — the channel is intrinsically tied to the ground-field continuity established earlier in this series. The antisymmetric channel emits nothing scalar at the octonion level — but after the contraction tower, its central output is exactly the quantum of action: $W = -\frac{1}{2}\{X^1, P_1\}$, the mediator regenerated as the relation between the surviving triples, central and scalar by the centrality theorem. One product, two scalar outputs of relations: the measurement pairing from the symmetric half, the $\hbar$ pairing from the antisymmetric half.

4. The measurement channel: three formalization rules

The identification of scalar emission with measurement must be handled so as not to damage the series' type III₁ architecture. Three rules, each with its justification, fix the treatment.

Rule 1: evaluation, never projection. The emission of a real number is formalized as evaluation through the Jordan/pairing channel — a state applied to an observable — not as projection onto an eigenstate. This is not merely prudent: in the type III₁ setting there is nothing for the projection picture to refer to — no minimal projections and no pure normal states exist, and by Connes–Størmer homogeneity all normal states are approximately unitarily equivalent. Events are irreducibly coarse; collapse-to-pure-outcome has no referent.

Rule 2: the pairing is the template of states, not a state. The pairing takes two algebra elements; a state takes one. They meet in exactly one way: every state on the finite core has the form $\omega(a) = \langle a, h \rangle$ for a positive density element h . The geometry supplies the pairing canonically; physics enters as h . The one forbidden choice is the geometry-only state — the pairing with no symmetry-breaking h — which is the trace, whose modular flow is trivial: it abolishes thermal time and with it the entire dynamical architecture.

Time exists because the physical state breaks the pairing's own symmetry.

Rule 3: outcome-production stays outside the formalism. The mathematics of this series constrains the *shape* of events (coarse, projectionless) but does not produce individual outcomes, probabilities, or the "from time to time" discreteness of measurement. That remains an open problem of quantum foundations, named here as outside scope rather than implicitly claimed as solved.

5. The operator-algebraic mirror

The split of §1 has an exact counterpart one stratum up. Alfsen and Shultz characterize when a Jordan-Banach algebra of observables is the self-adjoint part of a C*-algebra: precisely when it carries a *dynamical correspondence*, and then the associative product is reconstructed as $ab = a \circ b - i(a \star b)$ — the Jordan half plus the dynamical (antisymmetric) half. The full operator product is, there as here, Jordan-plus-flux; and the exceptional $J_3(\mathbb{O})$ is exactly the case admitting no dynamical correspondence — observables without time, as established elsewhere in this series. The octonion-level split and the operator-level split are one architecture at two strata: the symmetric half carries what is measured, the antisymmetric half carries what moves, and only where both exist does the associative world assemble.

6. Scope

This note does not address, and does not claim progress on, the measurement problem proper: no mechanism of outcome selection, no derivation of Born weights, no account of stochasticity is offered — Rule 3 is the formal registration of that boundary.

Proposition 1 is scoped to imaginary elements; on all of $\mathbb{O}$ the symmetric half is not purely scalar. The identification of the symmetric channel with physical measurement is an interpretive alignment — motivated by the Jordan tradition and consistent with the series' architecture — not a theorem.

References

- Companion papers: *J3(O): Construction, the Peirce Split, and Verified Frame Relativity via F4; The Central Element of Algebraic R-Flux; The Imaginary Octonions Are the Unique Non-Lie Malcev Algebra in the Hurwitz Chain; The Second Contraction: From Algebraic R-Flux to the Heisenberg Algebra, the Dilation Symmetry It Creates, and the SO(3) Remnant of G2; The Squeeze-Thermal State: A Type-I No-Go, the Scale-Line Construction, and the Type III₁ Verdict.*
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Created in collaboration with Claude (Anthropic). All conclusions and any errors remain the author's responsibility.